Lecture 8

Transmitter, Receiver and Noise

1

Topics Covered (Chapter 8–9)

- Transmitter Fundamentals
- Receiver Fundamentals
- Noise
- Noise Factor, Noise Figure

2

2

8-1: Transmitter Fundamentals

- A **radio transmitter** takes the information to be communicated and converts it into an electronic signal compatible with the communication medium.
- This process involves carrier generation, modulation, and power amplification. The signal is fed by wire, coaxial cable, or waveguide to an antenna that launches it into free space.
- Typical transmitter circuits include oscillators, amplifiers, frequency multipliers, and impedance matching networks.

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Transmitter

- The **transmitter** is the electronic unit that accepts the information signal to be transmitted and converts it into an RF signal capable of being transmitted over long distances.

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Transmitter

Every transmitter has four basic requirements:

1. It must generate a carrier signal of the correct frequency at a desired point in the spectrum.
2. It must provide some form of modulation that causes the information signal to modify the carrier signal.
3. It must provide sufficient power amplification to ensure that the signal level is high enough to carry over the desired distance.
4. It must provide circuits that match the impedance of the power amplifier to that of the antenna for maximum transfer of power.

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Transmitter Configurations

- The simplest transmitter is a single-transistor oscillator connected to an antenna.
- This form of transmitter can generate **continuous wave (CW)** transmissions.
- The oscillator generates a carrier and can be switched off and on by a telegraph key to produce the dots and dashes of the International Morse code.
- CW is rarely used today as the oscillator power is too low and the Morse code is nearly extinct.

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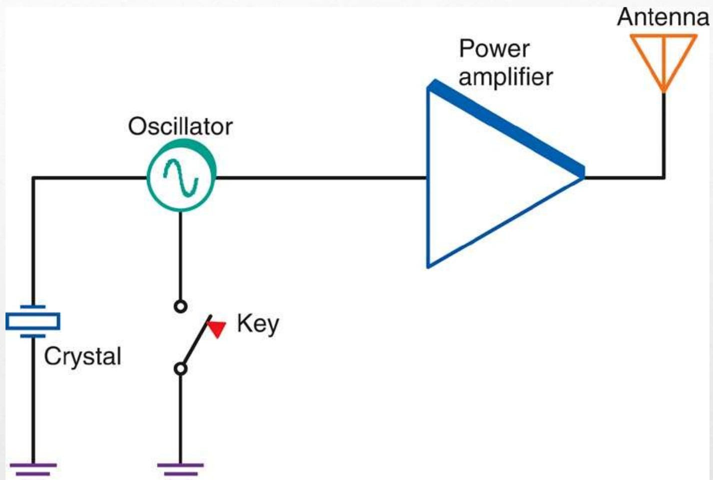

Figure 8-1: A more powerful CW transmitter.

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Transmitter Types

- High-Level Amplitude Modulated (AM) Transmitter

1. Oscillator generates the carrier frequency.
2. Carrier signal fed to buffer amplifier.
3. Signal then fed to driver amplifier.
4. Signal then fed to final amplifier.

8

8

Transmitter Types

- Low-Level Frequency Modulated (FM) Transmitter

1. Crystal oscillator generates the carrier signal.
2. Signal fed to buffer amplifier.
3. Applied to phase modulator.
4. Signal fed to frequency multiplier(s).
5. Signal fed to driver amplifier.
6. Signal fed to final amplifier.

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Transmitter Types

- Single-Sideband (SSB) Transmitter

1. Oscillator generates the carrier.
2. Carrier is fed to buffer amplifier.
3. Signal is applied to balanced modulator.
4. DSB signal fed to sideband filter to select upper or lower sideband.
5. SSB signal sent to mixer circuit.
6. Final carrier frequency fed to linear driver and power amplifiers.

10

10

9-1: Receiver Fundamentals

- In radio communication systems, the transmitted signal is very weak when it reaches the receiver, particularly when it has traveled over a long distance.
- The signal has also picked up noise of various kinds.
- Receivers must provide the sensitivity and selectivity that permit full recovery of the original signal.
- The radio receiver best suited to this task is known as the **superheterodyne receiver**.

11

11

Receiver

- A communication receiver must be able to identify and select a desired signal from the thousands of others present in the frequency spectrum (**selectivity**) and to provide sufficient amplification to recover the modulating signal (**sensitivity**).
- A receiver with good selectivity will isolate the desired signal and greatly attenuate other signals.
- A receiver with good sensitivity involves high circuit gain.

12

12

Receiver

Selectivity: *Q* and Bandwidth

- Selectivity in a receiver is obtained by using tuned circuits and/or filters.
- *LC* tuned circuits provide initial selectivity.
- Filters provide additional selectivity.
- By controlling the *Q* of a resonant circuit, you can set the desired selectivity.
- The optimum bandwidth is one that is wide enough to pass the signal and its sidebands but narrow enough to eliminate signals on adjacent frequencies.

13

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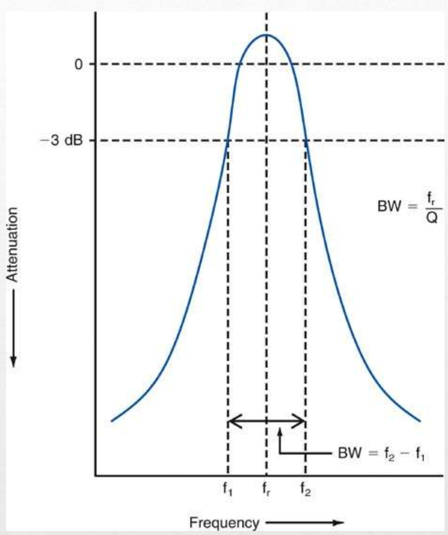

Figure 9-1: Selectivity curve of a tuned circuit.

14

14

Selectivity: Shape Factor

- The sides of a tuned circuit response curve are known as **skirts.**
- The steepness of the skirts, or the **skirt selectivity**, of a receiver is expressed as the **shape factor,** the ratio of the 60-dB down bandwidth to the 6-dB down bandwidth.
- The lower the shape factor, the steeper the skirts and the better the selectivity.

15

15

Sensitivity

- A communication receiver's sensitivity, or ability to pick up weak signals, is a function of overall gain, the factor by which an input signal is multiplied to produce the output signal.
- The higher the gain of a receiver, the better its sensitivity.
- The more gain that a receiver has, the smaller the input signal necessary to produce a desired level of output.
- High gain in receivers is obtained by using multiple amplification stages.

16

16

Sensitivity

- Another factor that affects the sensitivity of a receiver is the signal-to-noise (S/N) ratio (SNR).
- One method of expressing the sensitivity of a receiver is to establish the **minimum discernible signal (MDS).**
- The MDS is the input signal level that is approximately equal to the average internally generated noise value.
- This noise value is called the **noise floor** of the receiver.
- MDS is the amount of signal that would produce the same audio power output as the noise floor signal.

17

17

Basic Receiver Configuration

- The simplest radio receiver is a crystal set consisting of a tuned circuit, a diode (crystal) detector, and earphones.
- The tuned circuit provides the selectivity.
- The diode and a capacitor serve as an AM demodulator.
- The earphones reproduce the recovered audio signal.

18

18

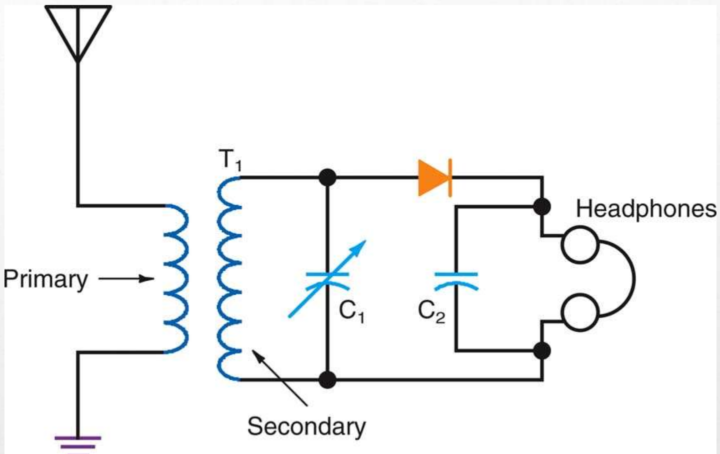

Figure 9-4: The simplest receiver—a crystal set.

19

19

9-5: Noise

- **Noise** is an electronic signal that gets added to a radio or information signal as it is transmitted from one place to another.
- It is not the same as interference from other information signals.

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20

9-5: Noise

- Noise is the static you hear in the speaker when you tune any AM or FM receiver to any position between stations. It is also the "snow" or "confetti" that is visible on a TV screen.
- The noise level in a system is proportional to temperature and bandwidth, the amount of current flowing in a component, the gain of the circuit, and the resistance of the circuit.

21

21

Signal-to-Noise Ratio

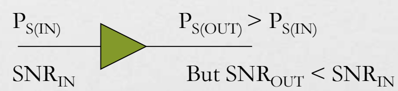
- The signal-to-noise (S/N) ratio indicates the relative strengths of the signal and the noise in a communication system.
- The stronger the signal and the weaker the noise, the higher the *S/N* ratio.
- The S/N ratio is a power ratio.
- SNR = P_{signal} / P_{Noise} or SNR (dB) = P_S(dBm) – P_N (dBm)

22

22

Signal to Noise Ratio

- Noise may cause an error at the receiver.
  - The error will depend on modulation type, signal bandwidth, signal power etc. Fundamentally depends on signal-to-noise ratio. The receiver error will decrease as the SNR increases.
- Amplification is not the SNR solution because noise is added in at the amplifier.



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23

External Noise

- External noise comes from sources over which we have little or no control, such as:
  - Industrial sources
    - motors, generators, manufactured equipment
  - Atmospheric sources
    - The naturally occurring electrical disturbances in the earth's atmosphere; atmospheric noise is also called **static**.
  - Space
    - The sun radiates a wide range of signals in a broad noise spectrum.

24

24

Internal Noise

- Electronic components in a receiver such as resistors, diodes, and transistors are major sources of **internal noise**. Types of internal noise include:
  - Thermal noise
  - Semiconductor noise
  - Intermodulation distortion

25

25

Thermal Noise

- Thermal noise or Johnson noise
  - Is generated by the random thermal motion of electrons, inside an electrical conductor, which happens regardless of any applied voltage.
  - Thermal noise is *white*, meaning that its power spectral density is nearly equal throughout the frequency spectrum.
  - A communication system affected by thermal noise is often modelled as an additive white Gaussian noise (AWGN) channel.
  - The root mean square (RMS) voltage due to thermal noise v_N generated in a resistance R (ohms) over bandwidth B (hertz), is given by

$$v_N = \sqrt{4kTBR}$$

where k is Boltzmann's constant (1.38×10^{-23} joules per kelvin) and T is the resistor's absolute temperature (kelvin).

26

26

Other Types of Noise

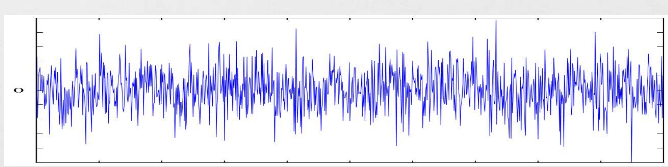
- Sky noise—refers to fluctuations in total power caused by variations in atmospheric emissivity and path length in timescales of order one second.
- Flicker noise—also known as *1/f noise*, is a signal or process with a frequency spectrum that falls off steadily into the higher frequencies, with a pink spectrum. It occurs in almost all electronic devices, and results from a variety of effects, though always related to a direct current.
- Shot noise—Shot noise in electronic devices consists of random fluctuations of the electric current in an electrical conductor, which are caused by the fact that the current is carried by electrons.

27

27

Additive White Gaussian Noise

- AWGN is commonly used to simulate background noise of a channel.
  - Additive – adds to the signal being transmitted.
  - White – noise that has a flat power spectral density. In other words, the signal contains equal power within a fixed bandwidth.
  - Gaussian - a Gaussian distribution of amplitude.
  - AWGN produces simple and tractable mathematical model which is useful for gaining insight into the underlying behaviour of a system.



28

28

9-5: Noise

- The noise quality of a receiver can be expressed in the following terms:
  - The **noise factor** is the ratio of the S/N power at the input to the S/N power at the output.
  - When the noise factor is expressed in decibels, it is called the **noise figure**.
  - Most of the noise produced in a device is thermal, which is directly proportional to temperature. Therefore, the term **noise temperature** (T_N) is used.
  - **SINAD** is the composite **si**gnal plus **n**oise **a**nd **d**istortion divided by noise and distortion contributed by the receiver.

29

29

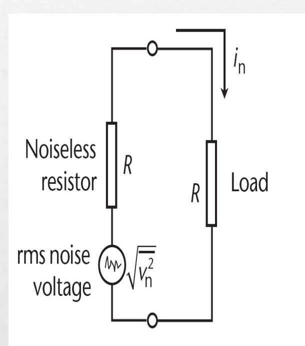
Available Noise Power

- Maximum noise power delivered to the load –

$$i_n = \frac{\sqrt{\overline{v_n^2}}}{2R} \qquad p_n = kTB$$

- The noise power per unit bandwidth (in W/Hz)

$$N_o = kT$$



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30

Equivalent Noise Temperature

- Equivalent noise temperature (T_e) is a parameter that indicates how noisy a circuit is.
- Equivalent noise temperature is always referred to the input of the circuit.
- A resistor connected to the input of a circuit would have to be raised to a temperature T_e to generate the same amount of thermal noise (measured at the output of the noiseless circuit) as the noisy circuit.
- At the output $P_N = GkT_eB$
- T_o = represents room temperature = 290k

31

31

Noise Factor and Noise Figure

- Noise Factor
  - A dimensionless quantity.
  - A figure of merit that compares the noise in a network to the noise in an ideal noiseless network.

$$F = \frac{SNR_{IN}}{SNR_{OUT}}$$

at a temperature of 290 K

- Noise Figure
  - Uses decibels to express the quantity.
  - $NF = 10\log_{10}F$ decibels

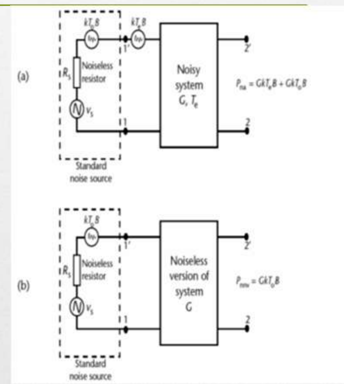
32

32

Noise Factor of a Single System

- F of a system is the ratio of the actual power output when the input is a standard noise source, to the noise power output that exists under the same conditions if the system is noiseless.
- Noisy system (noisy amplifier)
  - $P_{NA} = GkT_O B + GkT_e B$
- Noiseless system (no-noise version)
  - $P_{NNV} = GkT_O B$
- $F = P_{NA}/P_{NNV} = 1 + T_e/T_o$ and

$T_e = T_o(F\text{-}1) = 290(F\text{-}1)$



33

33

Noise Relationships

	Noise Figure NF [dB]	Noise Factor F_n [–]	Eq. noise temperature T_e [K]
Noise Figure NF [dB]	NF	$NF = 10\log_{10}(F_n)$	$NF = 10\log_{10}(T_e/T_0 + 1)$
Noise Factor F_n [–]	$F_n = 10^{NF/10}$	F_n	$F_n = T_e/T_0 + 1$
Eq. noise temperature T_e [K]	$T_e = (10^{NF/10} - 1)T_0$	$T_e = (F_n - 1)T_0$	T_e

$T_0 = 290$ K

34

34

NF of a Cascaded System

- Use the Friis formula for noise.

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \ldots$$

  - For a noiseless passive device, the noise factor is equal to the attenuation
    - F = 1/G = A. - otherwise treat as above.
    - A could be the insertion loss for a filter, coupler etc or the conversion loss of a mixer etc.
- Example - A Mini-Circuits MAR-1SM monolithic amplifier
  - Gain = 15 dB at 1000 MHz, NF = 5.5 dB, and there are other specifications.
  - Note that the noise figure is given and must be changed to a noise factor.

35

35

NF of a Cascaded System

- Noise has its greatest effect at the input to a receiver because that is the point at which the signal level is lowest.
- The noise performance of a receiver is determined in the first stage of the receiver, usually an RF amplifier or mixer.

36

36

How to Reduce Noise?

- Transpositions / twisting of pairs,
- Shielding,
  - Paired cables, coaxial cable, equipment boxes,
- Use only the necessary bandwidth,
- Provide power supply decoupling,
- Keep ground leads short as possible,
- Maintain separation between high and low-level signal leads,
- Use optical couplers

37

37